Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

[CGPA-3.32] in Cycle 1

IT13: ADBMS

Chapter 2:SQL

Practice Assignment Solution Template

Write a query and paste a screenshot of your output below the question.

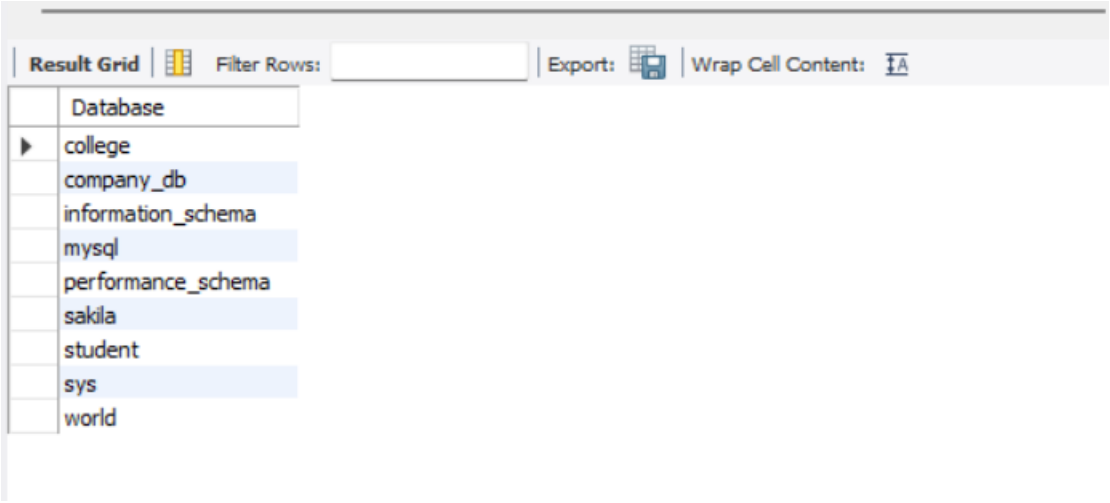
1. Create a database named CompanyDB.

   Solution:

   create database company_db;

   use company_db;

   show databases;



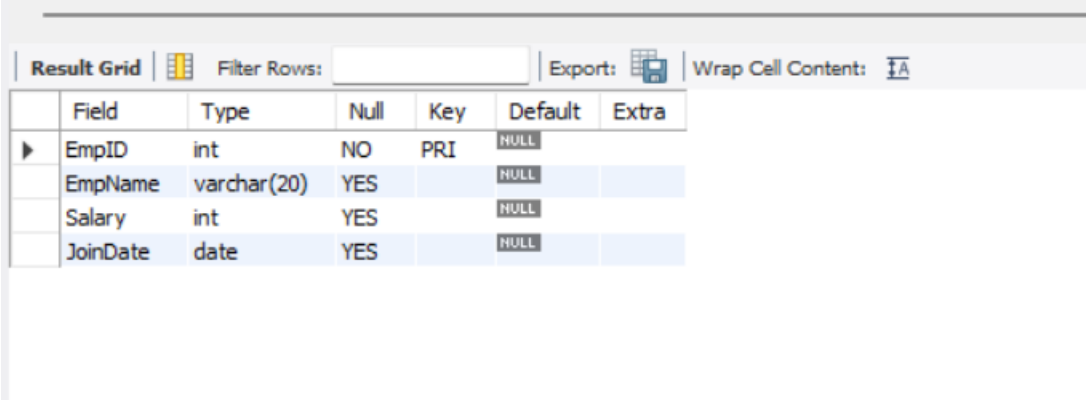
2. Create a table Employees with the following columns:

   Employees (EmpID (Integer, Primary Key), EmpName, Salary, JoinDate)

   Solution:

   create table Employees(EmpID int, EmpName varchar(20), Salary int, JoinDate date,Primary key(EmpId));

   desc Employees;



Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(20)	YES		NULL	
Salary	int	YES		NULL	
JoinDate	date	YES		NULL	

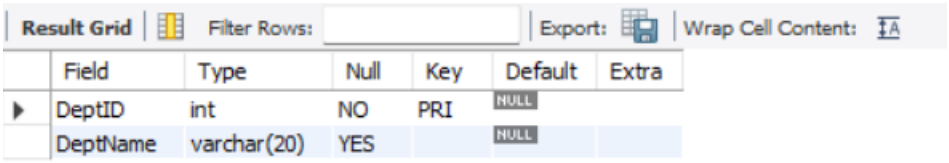
3. Create a table Departments with:

   Departments (DeptID (Integer, Primary Key), DeptName)

   Solution:

   create table Departments(DeptID int Primary Key, DeptName varchar(20));

   desc Departments;



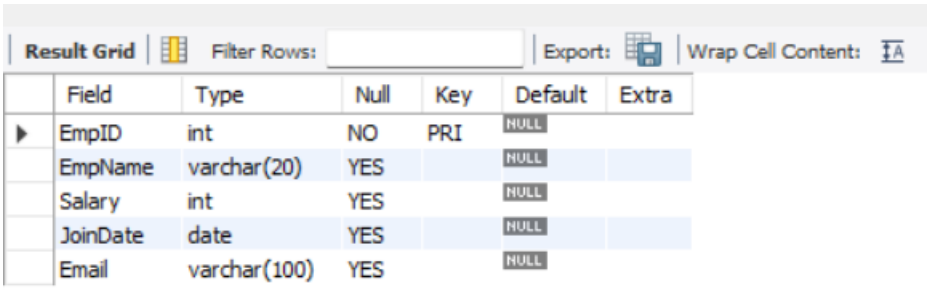
Field	Type	Null	Key	Default	Extra
DeptID	int	NO	PRI	NULL	
DeptName	varchar(20)	YES		NULL	

4. Add a column Email (VARCHAR(100)) to the Employees table.

   Solution:

   alter table Employees add column Email varchar(100);

   desc Employees;

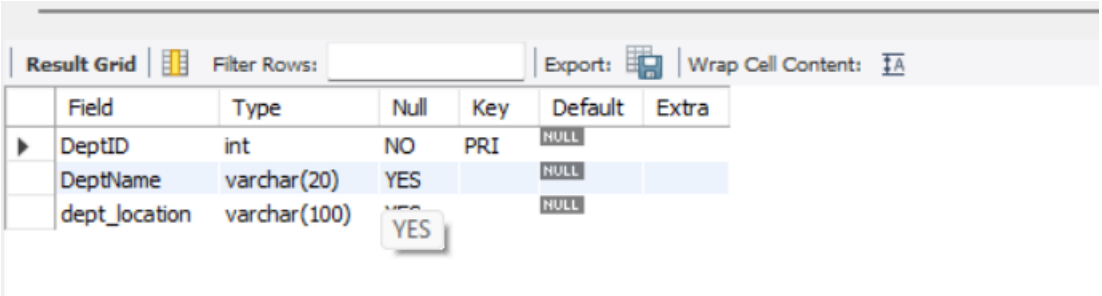


Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(20)	YES		NULL	
Salary	int	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	

5. Add a column dept_location to the Departments table.

   Solution:

   alter table Departments add column dept_location varchar(100);



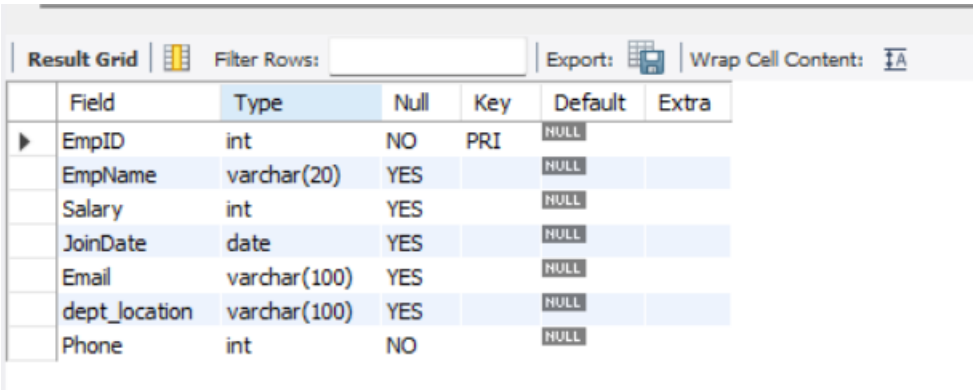
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
DeptID	int	NO	PRI	NULL	
DeptName	varchar(20)	YES		NULL	
dept_location	varchar(100)	YES		NULL	

6. Add a NOT NULL Constraint to the Phone column

   Solution:

   alter table Employees add column Phone int NOT NULL;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

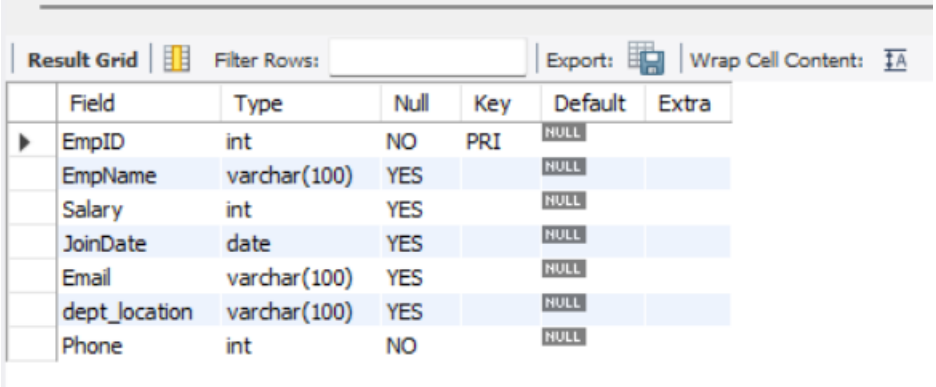
Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(20)	YES		NULL	
Salary	int	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	
Phone	int	NO		NULL	

7. Change the size of the EmpName column from VARCHAR (50) to VARCHAR (100).

   Solution:

   alter table Employees modify column EmpName varchar(50);

   alter table Employees modify column EmpName varchar(100);



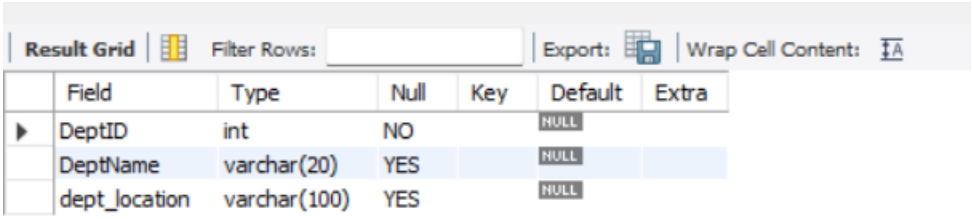
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(100)	YES		NULL	
Salary	int	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	
Phone	int	NO		NULL	

8. Drop the primary key Constraint from Departments

Solution:

alter table Departments drop primary key;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
DeptID	int	NO		NULL	
DeptName	varchar(20)	YES		NULL	
dept_location	varchar(100)	YES		NULL	

9. Create a foreign key constraint between two tables

Solution:

alter table Employees add DeptId int;

alter table Departments add primary key(DeptId);

alter table Employees add constraint fk_emp_dept foreign key (DeptId) references Departments(DeptId);

show create table Employees;

desc Employees;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(100)	YES		NULL	
Salary	int	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	
Phone	int	NO		NULL	
DeptId	int	YES	MUL	NULL	

10.Rename the column EmpName to EmployeeName.

Solution:

alter table Employees rename column EmpName to EmployeeName;

desc Employees;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(100)	YES		NULL	
Salary	int	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	
Phone	int	NO		NULL	
DeptId	int	YES	MUL	NULL	

11.Apply a CHECK constraint on the Age and Salary columns.

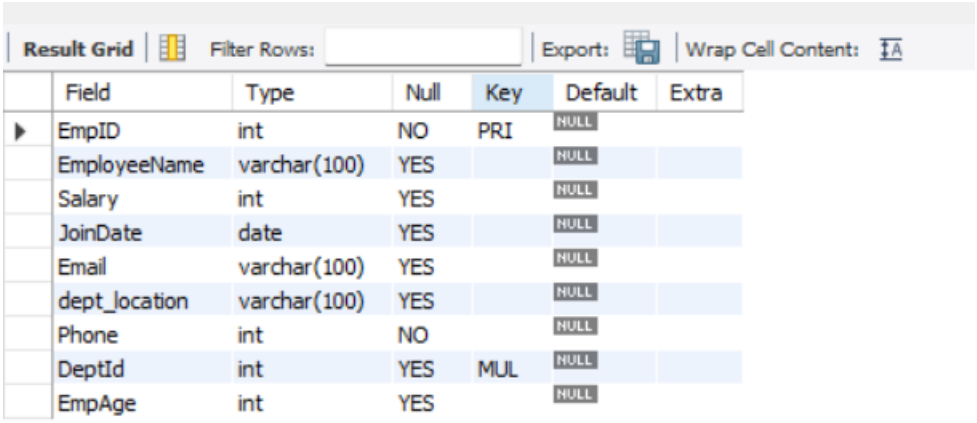
Solution:

alter table Employees add column EmpAge int;

alter table Employees add constraint chk_age CHECK (EmpAge >= 18);

alter table Employees add constraint chk_salary CHECK (Salary > 0);

desc Employees;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

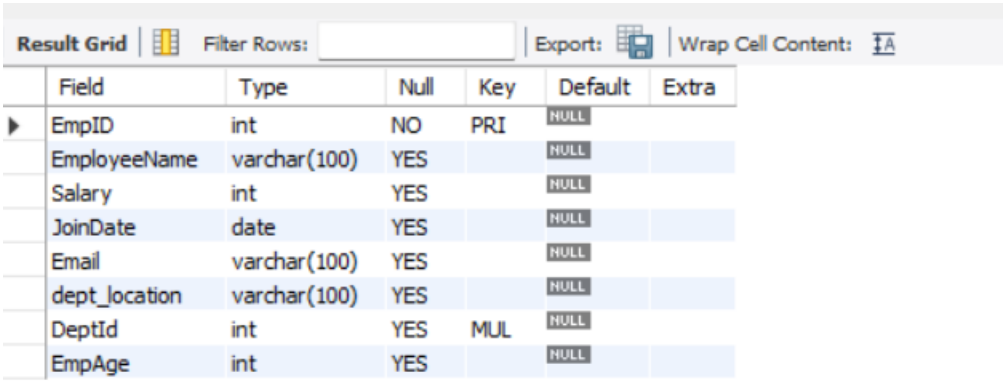
Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(100)	YES		NULL	
Salary	int	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	
Phone	int	NO		NULL	
DeptId	int	YES	MUL	NULL	
EmpAge	int	YES		NULL	

12. Remove the Phone column from the Employees table.

    Solution:

    alter table Employees drop column Phone;

    desc Employees;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(100)	YES		NULL	
Salary	int	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	
DeptId	int	YES	MUL	NULL	
EmpAge	int	YES		NULL	

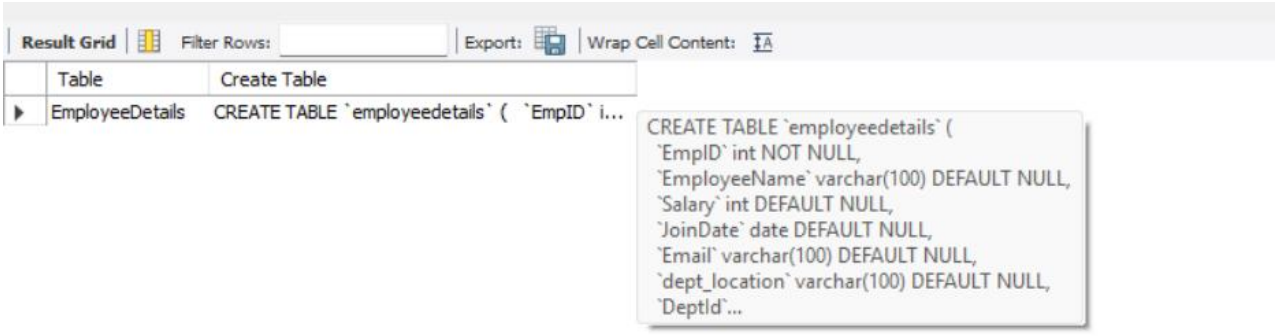
13. Rename the table Employees to EmployeeDetails.

    Solution:

    rename table Employees to EmployeeDetails;

    desc EmployeeDetails;

    show create table EmployeeDetails;



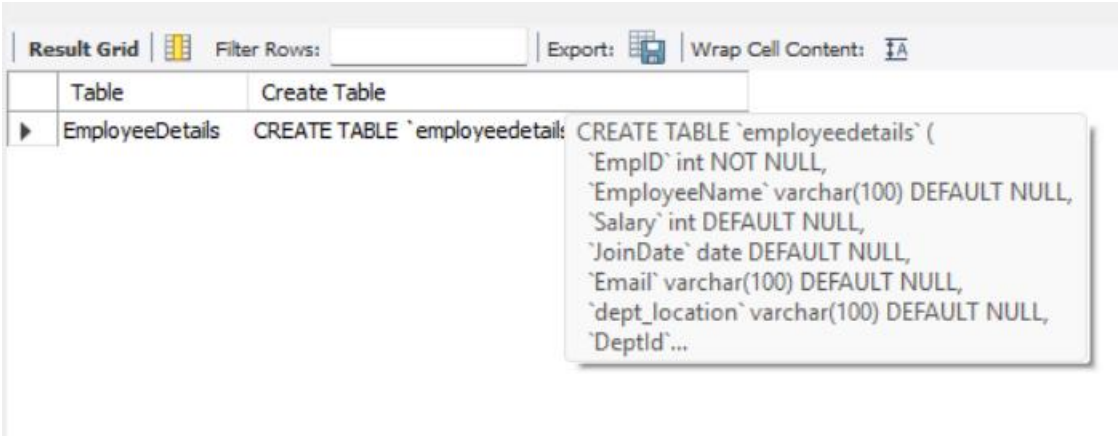
14. Drop the CHECK constraints.

    Solution:

    alter table EmployeeDetails drop check chk_age;

    alter table EmployeeDetails drop check chk_salary;

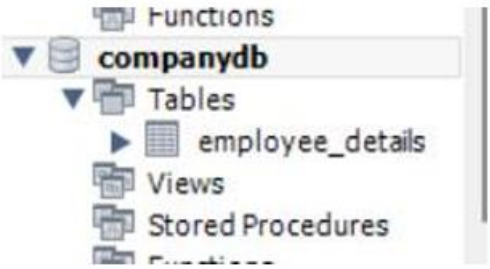
    desc EmployeeDetails;



15. Delete the Departments table permanently.

    Solution:- alter table employee_details drop foreign key fk_employees_departments;

    Drop table departments;



16. Drop the database.

    Solution:

    drop database company_db;

